

IR :-

- It holds the instruction i.e. Currently being executed.
- It hold the op in a Variable structure for the Control Circuit which generates the timing signal.

Operational steps :-

There are 10 operational step involved to execute a program or instruction.

The program stage in the memory and usually get the input through I/O devices.

Execution of the program starts when the program Counter is said to the first instruction of the program.

Content of the pc are transfer to MAR and READ Control signal is sent to the memory.

After the time required to access the memory the address is read out of the memory and loaded into MDR.

Now the Content of MDR are transfer to IR and the instruction is ready to be executed.

The first operand is fetched by sending its address to MAR and simultaneously initiate or READ Signal.

After one or two more cycle the ALU can perform the desired operation.

If the result of the operation is to be stored in the memory then result is send to MDR.

The address of the memory result is stored in MAR and right operation will be perform out.

The Content of pc will be in auto implimented mode and address of the next instruction will be stored in the pc.